

## CONTACT


Email
lorena.loremarcos@gmail.com

Portfolio
lorenamarcos.dev

LinkedIn
lorena-marcos-6516272b5

GitHub
github.com/Lorenadata

Location
Madrid, Spain


## TECH STACK

- R · tidyverse · ggplot2 · Quarto
- Python · pandas · numpy
- matplotlib · seaborn · plotly
- scikit-learn · XGBoost · statsmodels
- SQL · MySQL · PostgreSQL / PostGIS
- FastAPI · React · Docker · Git
- Claude API · Generative AI

## ML & STATISTICS

- Supervised & unsupervised learning
- Classification & regression (GLM)
- Leakage-safe evaluation · temporal splits
- Outlier detection (LOF)
- EDA · Statistical inference
- Survey data analysis (RB050)

## LANGUAGES

- Spanish — Native
- English — B2 Upper Intermediate

## MATHEMATICS

- Linear Algebra
- Discrete Mathematics
- Inferential Statistics

# Lorena Marcos

Data Scientist · Applied Data Science · UCM

Madrid · Looking for a Data Science / Data Analytics internship from February 2027

## PROFILE

Applied Data Science student at Universidad Complutense de Madrid who also builds products. I work with R and Python on real-world data, from official survey microdata to a 150k-record cancellation model, and publish every analysis as a reproducible report. I built and run RaceTrack, a full-stack platform in production, and my portfolio at lorenamarcos.dev. I use Claude and the Anthropic API to automate analytical work, and I check my own numbers: my export scripts caught and corrected errors in two of my published reports.

## PROJECTS

### RaceTrack — Platform for organising popular races in Spain

*Full-stack product · In production* 2026

- Registration, Stripe Connect payments and payouts, bibs with QR codes, live timing, results, communities and sponsorships.
- ~66,000 lines of code, 2,165 automated tests, 57 database migrations; geospatial search with PostGIS.

FastAPI · PostgreSQL + PostGIS · React · Stripe · Docker

### Cancellation Prediction Model — Ride-Sharing (NCR)

*Machine Learning · XGBoost · Classification* 2026

- Leakage-safe XGBoost model predicting customer cancellations on 150k bookings, with a temporal train/test split and logistic and no-skill baselines.
- Showed that booking-time features carry no signal (PR-AUC 0.071 vs 0.070) and that a seemingly strong model (PR-AUC 0.48) was learning an artefact of the synthetic data generator.
- Built on my data-quality audit of the same data: structural MNAR, duplicate IDs, IQR and LOF outliers.

R · XGBoost · testthat · Quarto · ~150k records

### Rental Market & Income Inequality in Spain

*Statistics · Official INE Microdata · ECV 2024* 2025

- Survey-weighted (RB050) analysis of tenants vs landlords: 46.2% of tenants are low-income vs 26.2% of non-tenants.
- Mean equivalised income, P10/P50/P90 percentiles and P90/P10 inequality ratio; reproducible report on GitHub Pages.

R · dplyr · ggplot2 · Hmisc · INE ECV 2024

### Child Poverty in Spain

*Statistical Inference · Official INE Microdata · ECV 2024* 2025

- Weighted AROPE rate of 34.7% for children under 16 vs 24.3% for the rest of the population, by household type and housing tenure.
- Fully reproducible pipeline published as an interactive HTML report on GitHub Pages.

R · Quarto · INE ECV 2024

### Fiscalis Database — Tax Management for Self-Employed

*SQL · Relational Database Design* 2025

- Relational model for a SaaS tax platform: profiles, activities, clients, invoices, expenses, filings and alerts.
- MySQL 8 schema (11 tables, 10 foreign keys) with seed data covering the full invoicing flow.

SQL · MySQL · Relational Design

## EDUCATION

### BSc in Applied Data Science

*Universidad Complutense de Madrid (UCM)* 2023 – present

- Relevant coursework: Machine Learning, Supervised & Unsupervised Learning, Time Series, Statistics, Data Mining, R, Python, SQL, Linear Algebra.

## EXPERIENCE

### Solar Energy Hackathon — Predictive Pricing Model

*Competition · Team work · < 48 hours* 2024

- Built a predictive model for photovoltaic energy prices in under 48 hours as part of a team, and presented it to a technical panel.

Predictive Modeling · Feature Engineering · Python · Teamwork